DAY 4 OF 30

The Lakehouse

Where most Fabric projects begin.
Files and tables in one place.

Files **Tables** **SQL endpoint**

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Swipe →

THE PROBLEM

Lakes and warehouses each solved half the problem.

1	**Data lake**	Takes anything, hard to query
2	**Warehouse**	Easy to query, strict on input
3	**So teams ran**	Both, and copied between them

Two systems, two copies, and **a pipeline whose only job is to move data between them**.

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Swipe →

THE SIMPLE VERSION

A warehouse that accepts unpacked boxes.

A normal warehouse only takes goods that are already sorted and labelled.

A Lakehouse takes the **boxes as they arrive**, and gives you **shelves** for whatever you have sorted.

✓ **Files**	Raw data in any format
✓ **Tables**	Sorted data, as Delta tables
✓ **Both**	In the same item, on OneLake

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Swipe →

HOW IT WORKS

Inside a Lakehouse

1. **Files**: land CSV, JSON, Parquet, images, anything
2. **Spark notebooks** clean and shape the data
3. **Tables**: the result, saved as Delta
4. **SQL analytics endpoint**: query tables with T-SQL
5. **Power BI** reads the same tables

One copy

The table Spark wrote is the table SQL and Power BI read. Nothing is exported.

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Swipe →

FOR ENGINEERS

What to know before you build

The SQL endpoint is read-only

You can query and create views, not insert or update. Data is written by Spark or by ingestion tools.

Only Delta shows up as a table

A folder of plain Parquet or CSV stays in Files until you load it into a Delta table.

Schemas organise tables

New lakehouses created in the portal have schemas enabled, so group tables instead of one flat list.

Tables need care

Many small writes create many small files. Plan compaction from the start.

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Swipe →

THE HONEST PART

Is a Lakehouse the right start?

GOOD FIT

Your team is comfortable with Spark or Python

Data arrives as files, semi-structured or large

You want one store for engineering and BI

THINK TWICE

The team only knows SQL

You need multi-table transactions in T-SQL

Everything is small and already relational

> **The catch:** because the endpoint is read-only, SQL-first teams often feel stuck on day two. Decide who writes the data, and with what, before choosing.

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Swipe →

FROM REAL WORK

A Lakehouse on a real project

A people-analytics platform that replaced CSV files passed around by hand.

1. **Raw files** landed in the Lakehouse through pipelines
2. **PySpark** shaped them into Bronze, Silver and Gold tables
3. **Semantic models** sat on the Gold tables
4. **Row-level security** controlled who saw which rows

One item held the raw files **and** the reporting tables.

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Swipe →

SAVE THIS

The Lakehouse in five lines

1. **Files** for raw data, **Tables** for clean data
2. **Tables are Delta**, stored once in OneLake
3. **The SQL endpoint** reads, it does not write
4. **Power BI** uses the same tables directly
5. **Small files** are the first thing to watch

Tomorrow: Lakehouse or Warehouse. The question every Fabric team asks in week one.

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Follow for Day 5